PHYSICS

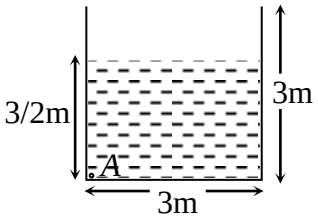
SECTION – I

Single Correct Choice Type

This section contains 8 single correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

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1. A rectangular container of height and width 3m each is filled with an ideal liquid upto a height of $\frac{3}{2}$m. This container is pulled horizontally by an acceleration equal to g. The gauge pressure at point A as shown in figure is P_1. If the container is completely filled with the liquid and closed from top and then pulled horizontally by same acceleration, the gauge pressure at same point is P_2. Then $\frac{P_1}{P_2}$ is



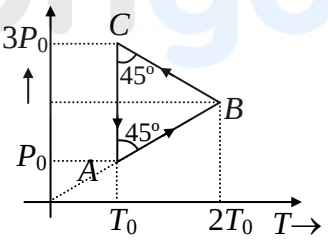
(a) 1 (b) $\frac{1}{3}$ (c) $\frac{1}{4}$ (d) $\frac{1}{2}$

2. A steel wire with cross section 3 cm^2 has elastic limit 2.4×10^8pa. The maximum upward acceleration that can be given to a 1200 kg elevator supported by this cable if the stress is not to exceed 1/3rd of the elastic limit is

(a) $9ms^{-2}$ (b) $10ms^{-2}$ (c) $11ms^{-2}$ (d) $12ms^{-2}$

3. An ideal monoatomic gas is undergoing a cyclic process as shown in the figure. Then choose the correct statement

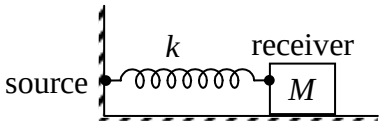
(a) work done by the gas in process CA is negative
(b) heat is rejected by the system in process BC
(c) work done by the gas in the cyclic process is positive
(d) internal energy of the gas decreases in process AB



4. A satellite is projected with velocity $\sqrt{\frac{5}{6}}$ times its escape speed from earth's surface. The initial velocity of the satellite is parallel to the surface of earth. The maximum distance of the satellite from the center of earth will be

(a) $3R$ (b) $4R$ (c) $5R$ (d) $8R$

5. In the given arrangement a source is placed at the fixed end of the spring with spring constant 100 N/m and the receiver is attached to the moving end of the spring, which inturn is connected to a box of mass 4kg, which is oscillating horizontally with amplitude of oscillation 2*m*. If the source emits a frequency of 990 Hz then (speed of sound = 330m/s)



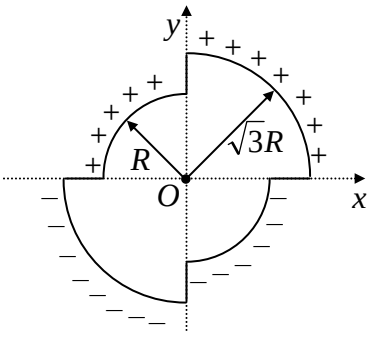
(a) maximum frequency received by the receiver is 990Hz

(b) minimum frequency received by the receiver is 990Hz

(c) frequency band width received by the receiver is 960Hz – 1020Hz

(d) maximum and minimum frequencies are received at the extreme positions.

6. In the figure there are four arcs carrying positive or negative charge all of them have charge density λ, then correct alternative is



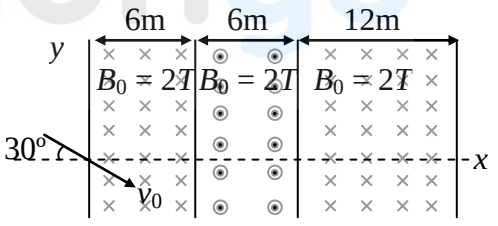
(a) the net dipole moment for the given charge distribution is $4\sqrt{5}\lambda R^2$

(b) the resultant electric field at the center and at any point on the axis passing through point *O* and perpendicular to the plane of arcs are opposite in direction

(c) if a uniform electric field is switched on perpendicular to the plane then the charge distribution starts rotating about x-axis

(d) potential at the center of the given charge distribution is non zero.

7. A particle having charge 1C and mass 1kg enters a region having uniform magnetic field of strength '2*T*' with a speed of 12 m/s, as shown in figure, then the correct statement is



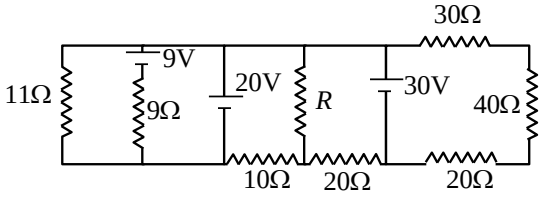
(a) the time for which the charge particle remains in magnetic field is 8.34s

(b) the velocity of charged particle becomes parallel to x-axis 6 times during its motion

(c) the shortest distance between the point where the charge particle enters the uniform magnetic field and the point where it emerges out is 6m

(d) the deviation of the charge particle when it emerges out of the magnetic field is '$2\pi/3$' rad.

8. In the given figure, the value of R so that power generated in R will be maximum is



(a) $\frac{45}{4}\Omega$ (b) $20\ \Omega$

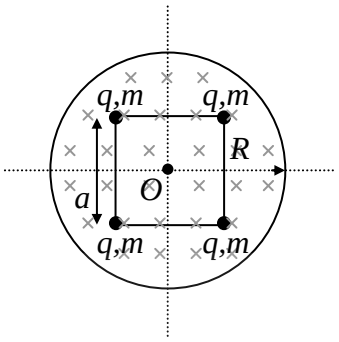
(c) $10\ \Omega$ (d) $\frac{20}{3}\Omega$

SECTION – II
Multiple Correct Choice Type

This section contains 4 multiple correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONE OR MORE** is/ are correct.

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9. Four identical charge particles each of mass 0.1kg and charge $2C$ connected to each other via massless non-conducting rods of equal length. The whole arrangement is placed in a cylindrical region carrying a uniform magnetic field as shown in the figure ($B_0 = 4T, a = 1m$). Suddenly the magnetic field is switched off. Then choose the correct statement(s)



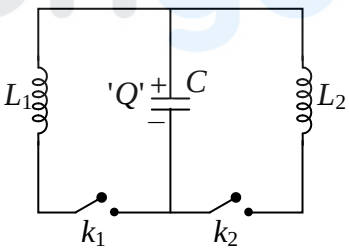
(a) the angular momentum of the system is 8 Nm/s

(b) the angular momentum of the system is $8\sqrt{2}$ Nm/s

(c) Angular velocity of the system is 40 rad/s

(d) the magnetic field at point O after the magnetic field is switched off is non zero.

10. The given arrangement carries a capacitor with capacitance 40mF and two inductors $L_1 = 25H$ and $L_2 = 100H$. If the capacitor initially carries a charge of 10 mC, then



(a) the maximum current through the inductor L_1 when key k_1 is closed is 20 mA

(b) the maximum current through the inductor L_2 when key k_2 is closed is 5mA

(c) the maximum current through inductor L_2 when both the keys are closed is $\sqrt{5}$ mA

(d) time period of oscillation of charge is minimum when both the keys are closed

11. Two identical point particles *A* and *B* are placed in front of a concave mirror of focal length 20cm, at distances 10 cm and 30 cm respectively. The particles oscillate perpendicular to the principal axis, such that the displacement equation for both the particles is given by $Y_A = Y_B = 0.1\sin(\pi t)$. The correct statement(s) is/are

(a) displacement equation of the image of particle *A* is $Y_A' = 0.2\sin(\pi t + \pi/2)$

(b) displacement equation of the image of particle *B* is $Y_B' = 0.3\sin(\pi t + \pi)$

(c) time period of oscillation of both the images is 2s.

(d) relative vertical displacement of image of A with respect to image of B is $Y_{AB}' = 0.4\sin(\pi t)$

12. With reference to the Young's double slit experiment, the correct statement(s) is/are

(a) if the separation between the screen and slits is increases, then the fringes inflate

(b) if the separation between the two slits in increased, then number of fringes on the screen decreases

(c) if the whole arrangement is immersed in a medium with refractive index μ, then the number of fringes on the screen will increase

(d) if the slits are illuminated with white light then the central maxima of the fringe pattern will also be white in colour

SECTION–III
Comprehension Type

This section contains 1 group of questions. Each group has 3 multiple choice questions based on a paragraph. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

If two deuterium nuclei get close enough together, the attraction of the strong nuclear force will fuse them to make as isotope of helium. The process will release a vast amount of energy. The range of nuclear force is 10^{-15} m. This is the principle behind the nuclear fusion reactor. The deuterium nuclei moves so fast hence it is not possible to contain them by physical walls. Therefore, they are confined magnetically.

13. How fast two nuclei shall come to have a head on collision to fuse?

(a) 1.2×10^6 ms^{-1} (b) 2.6×10^6 ms^{-1} (c) 3.8×10^6 ms^{-1} (d) 8.3×10^6 ms^{-1}

14. The strength of magnetic field required to make deuterium nuclei moving with this speed, to be confined in a circle of diameter 2.5m

(a) 134 mT (b) 129 mT (c) 139 mT (d) 116 mT

15. The following is the process of formation of the Helium nuclei from the nuclear fusion of deuterium nuclei

$${}_1^2H + {}_1^2H \rightarrow {}_1^3H + {}_1^1H + \Delta E_1$$

$${}_1^3H + {}_1^2H \rightarrow {}_2^4He + {}_0^1n + \Delta E_2$$

[$m({}_1^2H) = 2.014102$ a.m.u, $m({}_1^3H) = 3.016049$ a.m.u, $m({}_2^4He) = 4.002603$ a.m.u, $m({}_0^1n) = 1.008665$ a.m.u, $m({}_1^1H) = 1.007825$ a.m.u] then $\Delta E_1 + \Delta E_2$ is

(a) 12.3 MeV (b) 21.6 MeV (c) 17.7 MeV (d) 25.6 MeV

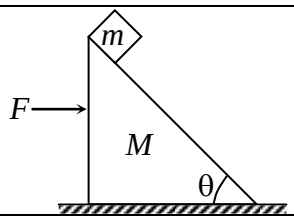
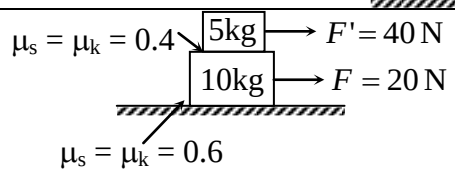
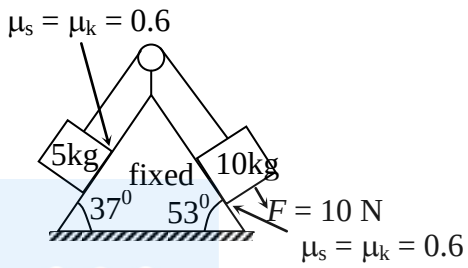
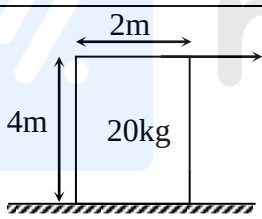
SECTION–IV
Matrix–Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled 1, 2, 3, 4 and 5. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

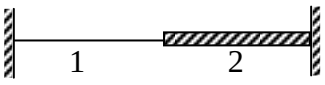
If the correct matches are A – 1, 4 and 5; B – 2 and 3; C – 1 and 2; and D – 4 and 5; then the correct darkening of bubbles will look like the following.

	1	2	3	4	5
A	●	○	○	●	●
B	○	●	●	○	○
C	●	●	○	○	○
D	○	○	○	●	●

1. In column I some arrangements of the objects are given. F represents the external constant force. In column II the possible variation of physical quantities are given. Match column I with column II ($\sin 37^0 = \frac{3}{5}$, $\cos 37^0 = \frac{4}{5}$)

Column I	**Column II**
(A) All surfaces are frictionless and $F = mg\sin\theta\cos\theta$ 	**1.** The centre of mass of the system accelerates in horizontal direction.
(B) 	**2.** The centre of mass of the system accelerates in vertical direction
(C) 	**3.** The object on which F is applied remains in translatory equilibrium
(D)  The box has uniform mass distribution	**4.** Kinetic energy of the system increases **5.** The friction force acting at all contact surfaces is equal to the corresponding limiting value of static friction.

2. Two strings 1 and 2 joined end to end and are fastened between two rigid supports under same tension. Mass per unit length of the strings are μ_1 and μ_2 respectively. A wave given as $y = A\sin(kx \pm \omega t)$ is generated in one of the string. A_r and A_t represents respectively the amplitudes of reflected and transmitted wave respectively from the interface. Match column I with column II.

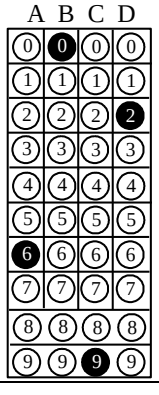


Column I	Column II
(A) $\mu_1 = 4\mu_2$ and incident wave is generated in string 1.	**1.** $A_r = \frac{A}{3}$ or $A_r = A$
(B) $\mu_1 = 4\mu_2$ and incident wave is generated in string 2.	**2.** $A_t = \frac{2A}{3}$ or $A_t = \frac{4A}{3}$
(C) $\mu_1 = \mu_2$ and incident wave is generated in string 1.	**3.** In either string, there will be a standing wave and a component of traveling wave.
(D) $\mu_1 << \mu_2$ and incident wave is generated in string 1.	**4.** In either string, there will be a standing wave. **5.** The energy of the system remains constant.

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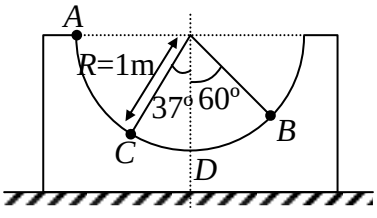
SECTION–V
Integer Answer Type

This section contains 4 questions. The answer to each of the questions is a **single digit integer,** ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers A, B, C and D (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the following:



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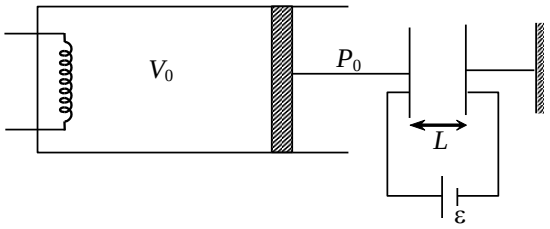
1. In the given arrangement there is a wedge of mass 10kg and a particle of mass 2kg, the friction co-efficient between the wedge and particle is 0.5, and there is no friction between the ground and wedge. When the particle is released from point *A*, it reaches point *B* on the wedge and comes rest momentarily, then it starts moving from point *B* and comes to rest at point *C*, momentarily if, then it starts from point *C* and finally comes to rest at point *D*. Calculate the magnitude of work done by friction force when it moves from point *B* and comes to momentary rest at point *C*?



Space for work

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2. An ideal gas is enclosed in a cylinder fitted with a frictionless piston. The piston is connected with a light rod to one plate of capacitor whose other plate is fixed as shown. Initially volume of the gas inside the cylinder is V_0. Atmospheric pressure is P_0, separation between the plates is L, area of the piston as well as of the capacitor plates is A and emf of battery is ε. A heater supplies heat to the gas so that pressure of the gas is given as $P = P_0 - \frac{n\varepsilon_0\varepsilon^2}{L^2}$, when piston is displaced by a distance $\frac{L}{2}$. Find value of n.



Space for work

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3. A man in a boat wishes to cross a 400 m wide river flowing at the rate of 2 m/s. Speed of boat in still water is 4 m/s. The person starts to row the boat perpendicular to the river flow from one bank of the river. When he reaches in the middle of the river. In order to reach the point directly opposite to the starting point on the opposite bank, he starts to row the boat at an angle θ with the vertical. The value of tan θ is $\frac{4}{n}$. Find value of *n*.

Space for work

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4. A tank has a hole area of cross-section *A*. Water from a pipe of inner cross-section 2*A* is entering the tank with a velocity of 5 m/s. calculate the height upto which the water will be filled inside the tank at steady state?

Space for work

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